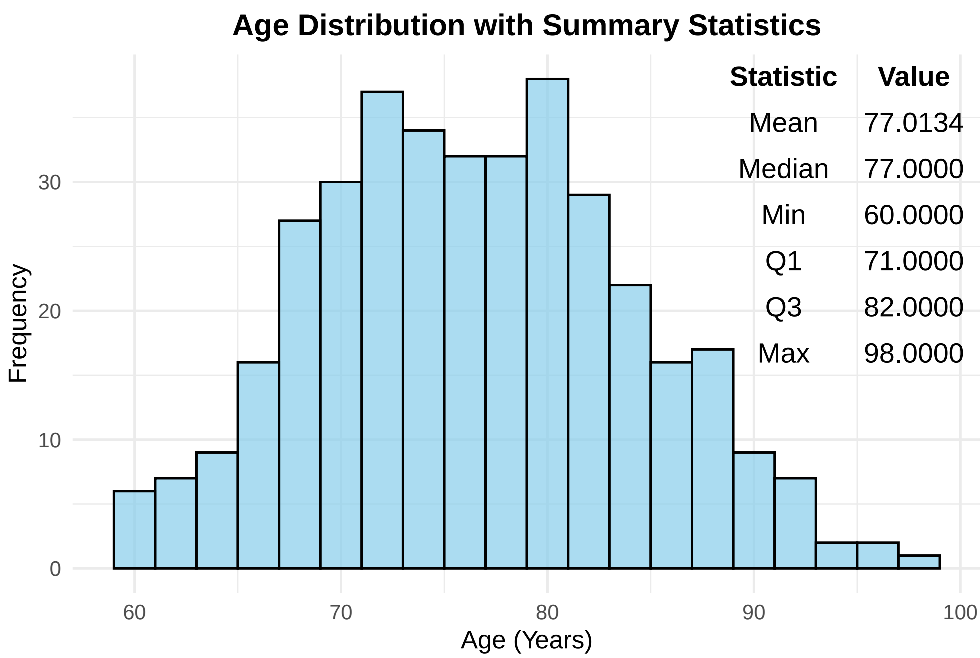
Analysis of OASIS Alzheimer's Detection Dataset

The Oasis data set consists of 373 scans, of patients aged 60 to 96, who received an MRI scan on two or more visits, each visit separated by at least a year. The cognitive status (demented/non demented) was documented throughout the study.

Demographic Information of Patients

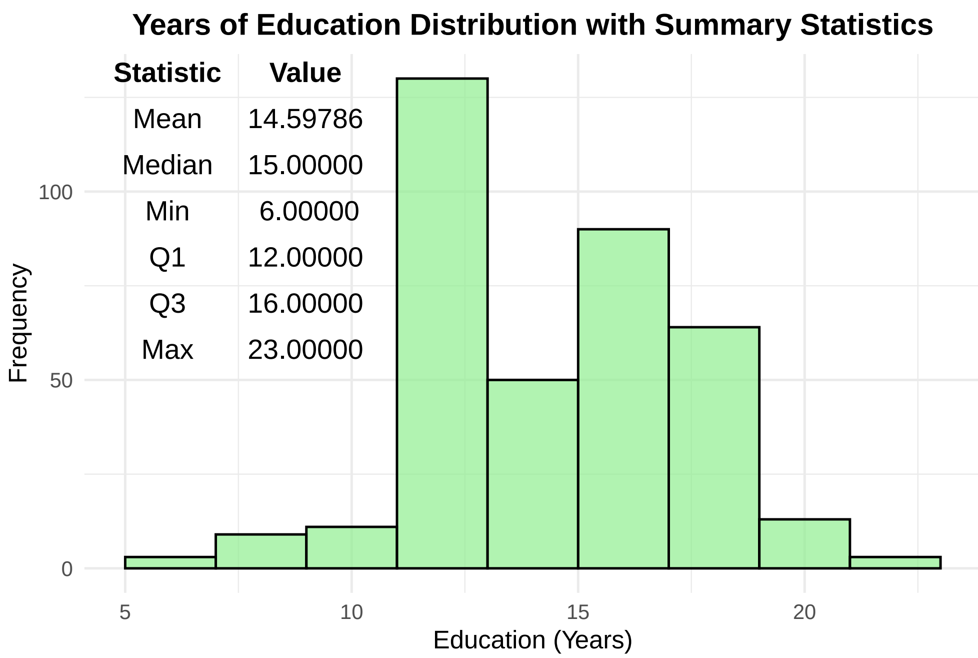
Age

The patients in this dataset were in the age range of 60-96. Below is a histogram of the distribution of ages, along with the mean, min, max, and quartile data. The ages of participants follow a relatively normal distribution, with very similar mean and median values of 77.01 years and 77 years, respectively.



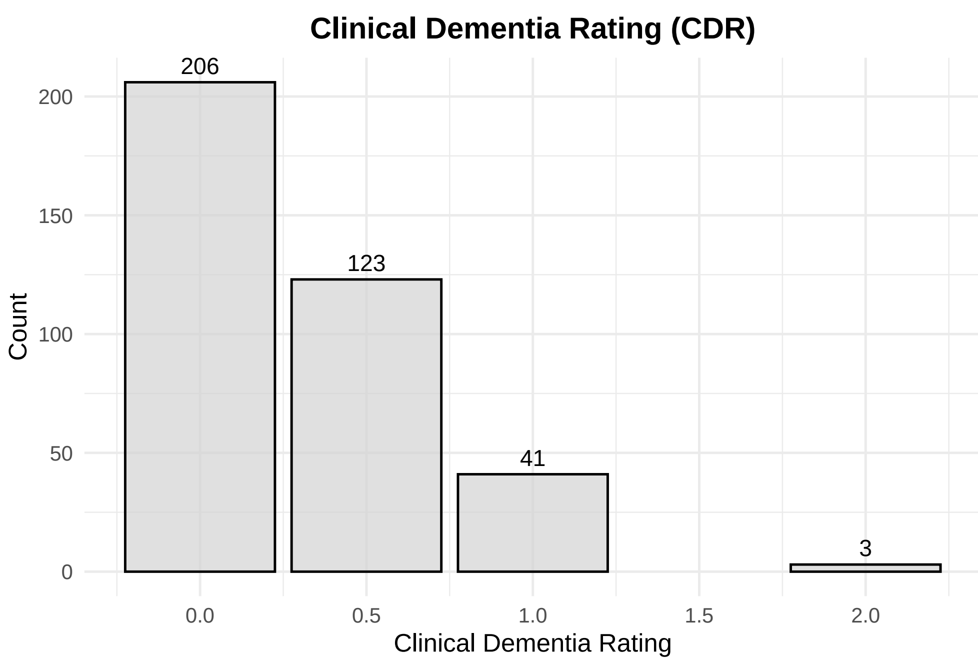
Years of Education

The years of edcuation each patient received was noted, with a minimum of 6 years and a maximum of 23 years. The histogram with associated statistics is shown below. The mode of this dataset is 12 years, indicating the participant finished high school, as shown by the peak on the histogram, with the second most common education length being 16, inidicating the particpant finished college.



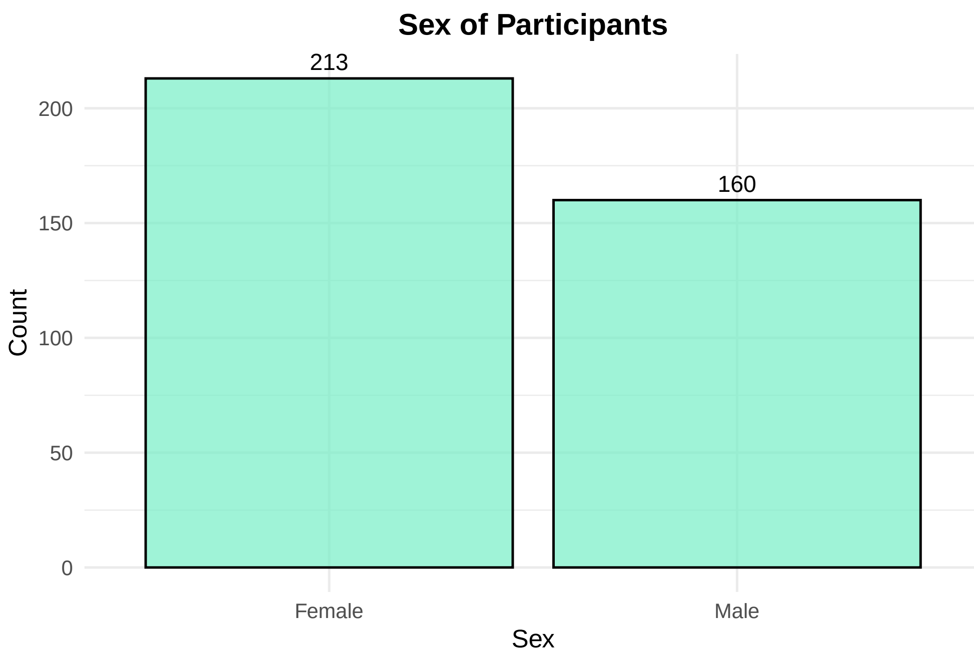
Clinical Dementia Rating

The clinical dementia rating, or CDR, is a 0-3 scale, with severe cognitive impairment being a 3. As shown by the histogram below, the values of 1.5-3 are underrepresented in this dataset.



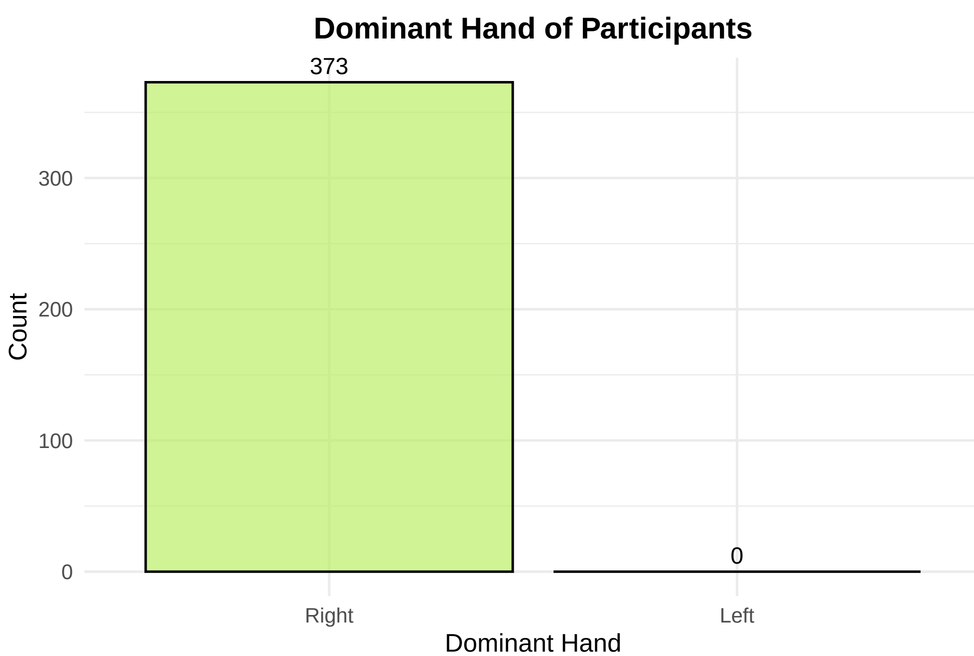
Sex of Patient

This study had more female participants than male participants, with 57% of patients being female.



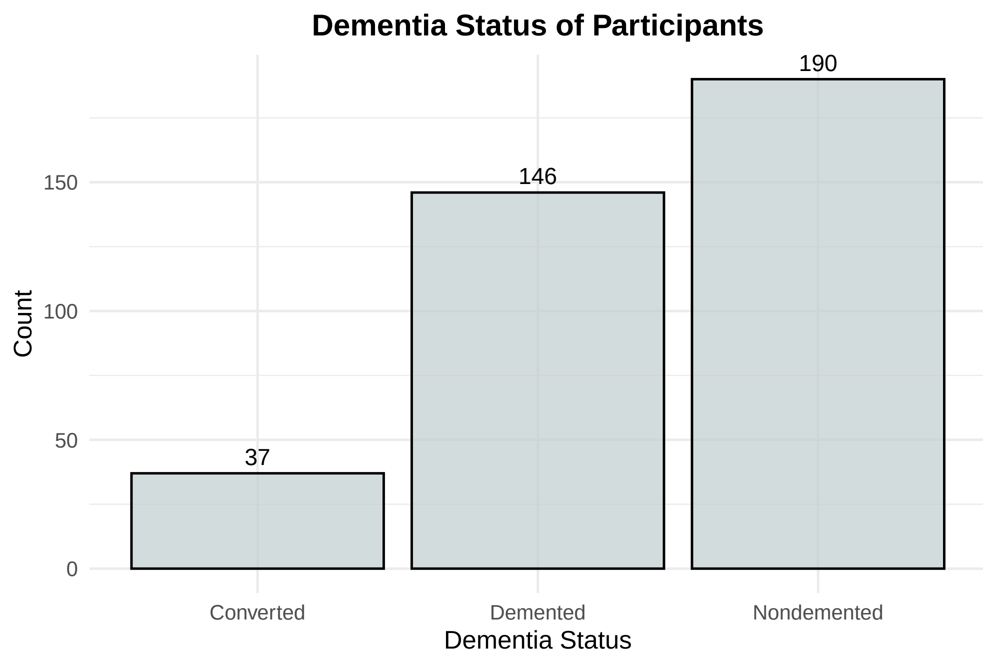
Dominant Hand

This study noted the dominant hand of participants, which was interesting because all participants were right handed.



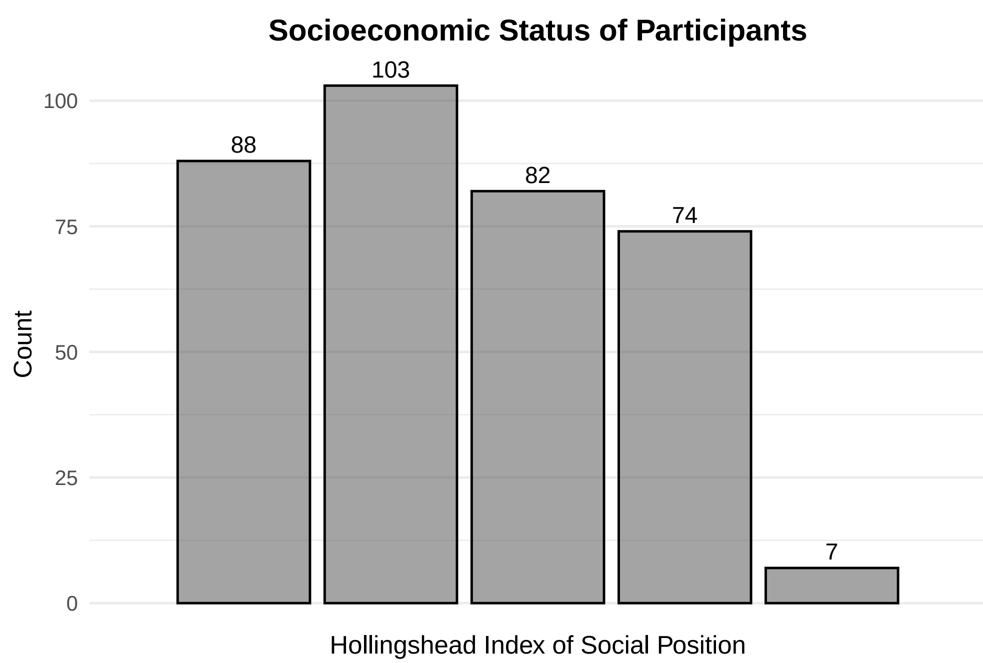
Dementia Status

The study noted the dementia status of each patient as nondemented, demented, or converted, which would mean that they were considered nondemented at the start of the study and later considered demented. The study included 190 nondemented patients, 146 patients diagnosed with dementia, and 37 patients whose status changed from nondemented to demented over the course of the study.



Socioeconomic Status

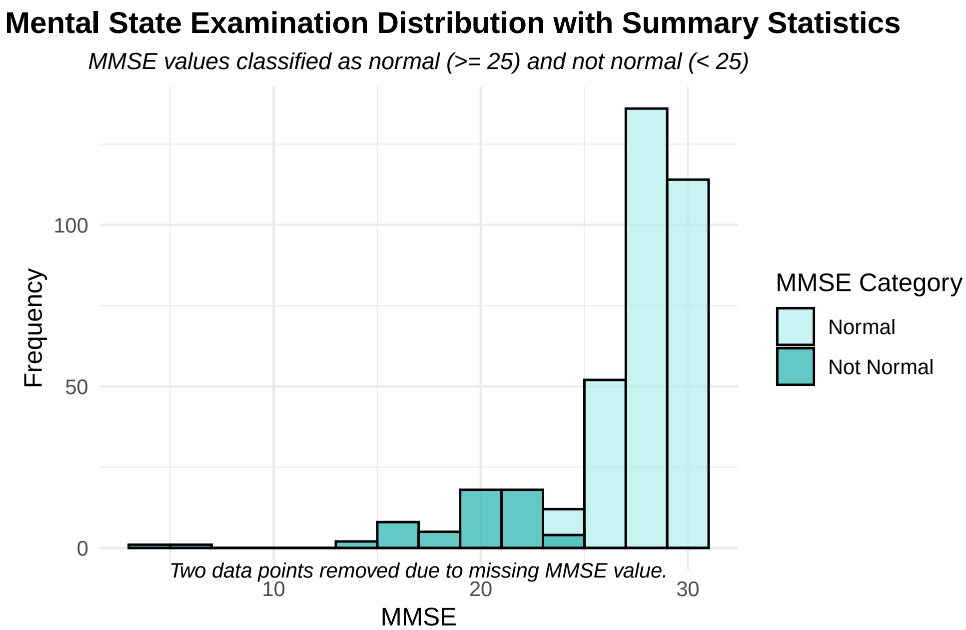
The socioeconomic status of participants is scored by the Hollingshead Index of Social Position (HISP), where 1 is the highest status and 5 is the lowest. There were 19 participants who did not have their socioeconomic status listed. There were very few patients of the lowest socioeconomic status. The other groups were relatively similar in size, with the largest group being those who scored a 2 on the HISP.



Raw Data Figures

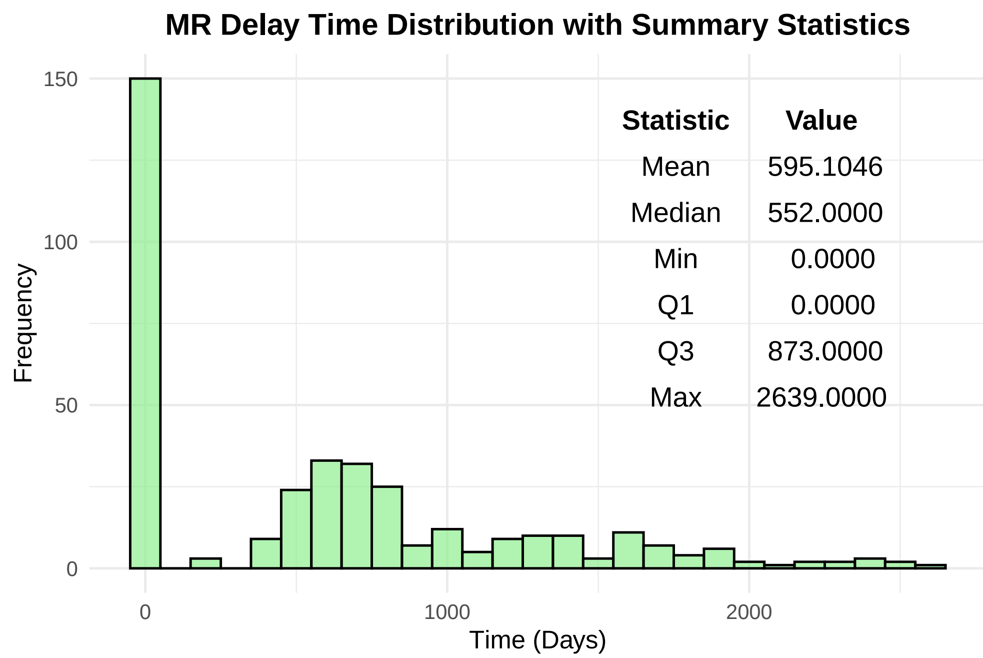
Mini Mental State Examination

The mini mental state examination (MMSE) assesses cognitive impairment, where lower scores indicate more significant cognitive impairment. A score of 25 or greater is considered normal, and the score range from 0-30.



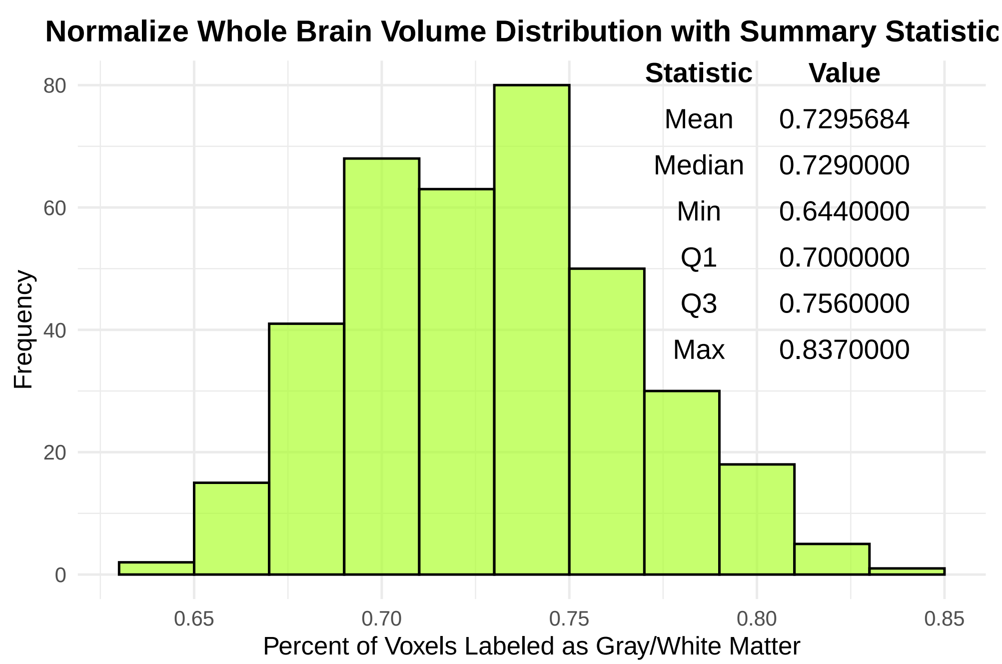
MR Delay Time

The MR delay time notes the days between each MRI the patient undergoes. The peak at zero indicates the first MRI the patient underwent. Excluding 0, the MR delay ranges from 182 to 2639 days.



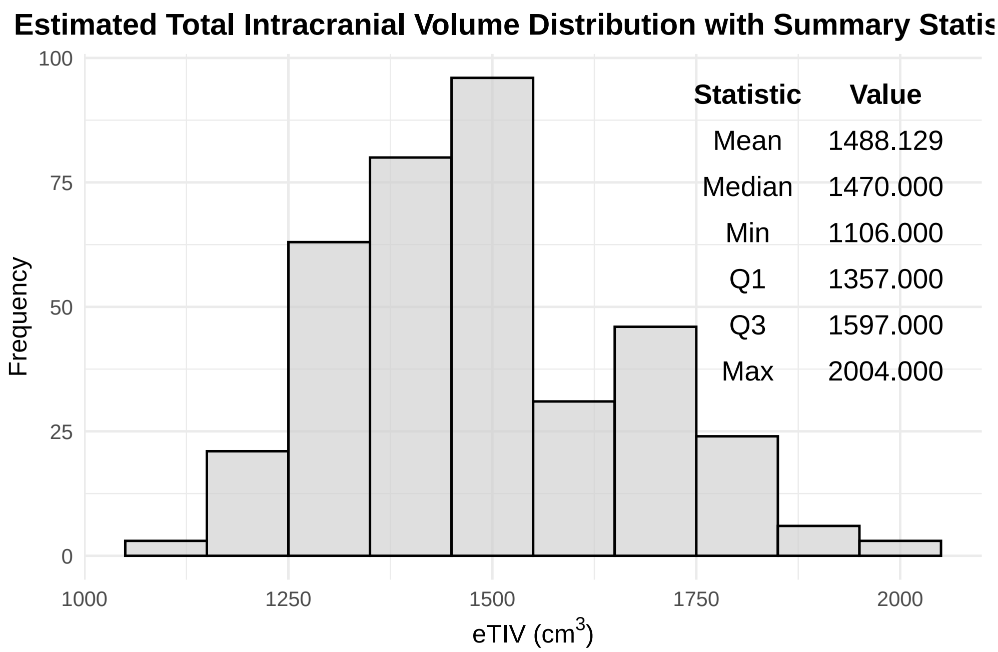
Normalized Whole Brain Volume

The normalized whole brain volume is the measurement of the total brain volume after adjusting it for individual head size. As shown in the histogram and statistics chart below, it follows a relatively normal distribution and the mean and median are very similar.



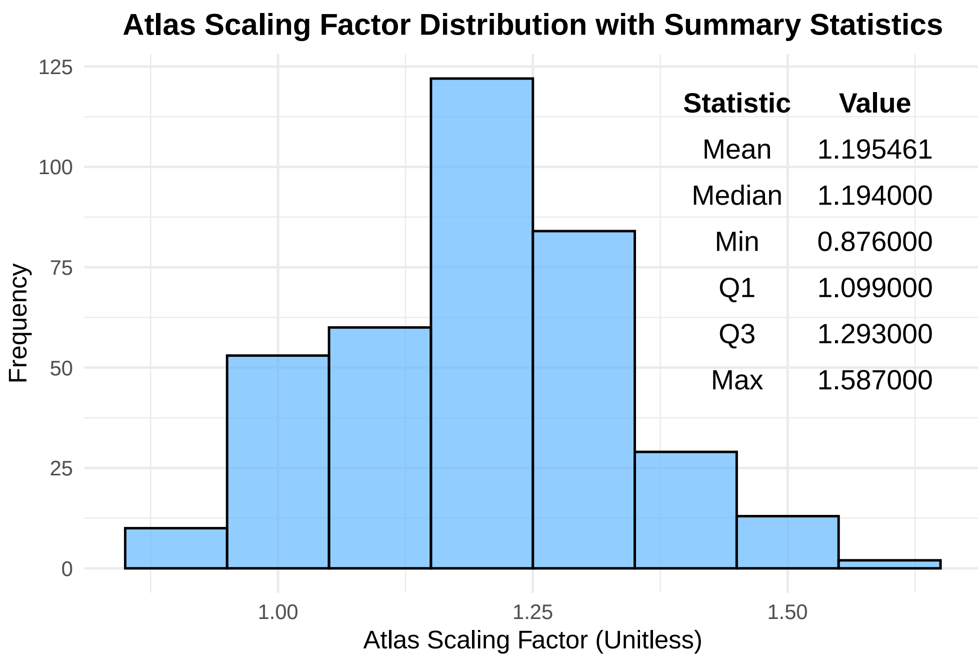
Estimated Total Intracranial Volume

The estimated total intracranial volume is the estimated volume of the cranial cavity as outlined by the supratentorial dura matter. This measure stays consistant with age, and is used to correct of head size variation across subjects in the study when computing the normalized whole brain volume. This data is skewed right, as shown by difference in the mean, 1488 cm^3, and the median, 1470 cm^3.



Atlas Scaling Factor

The Altas Scaling Factor is used standardise brain measurements across patients while accounting for differences in head size and shape, allowing the computation of the normalized whole brain volume. This factor follows a relatively normal distribution, as shown by the histogram below. There is not significant skewness in the data, represented by the very similar median and mean.



Analysis

Age and Dementia Group

Though these patients are 60 and older, it was expected that the demented group would be older than the non-demented group, especially because some patients changed groups from non-demented to demented over the course of the study. However, it was found that the non-demented and demented groups were not significantly different in age, but the converted group was significantly different than the demented group with a p-value of 0.034, through the Tukey's HSD test.

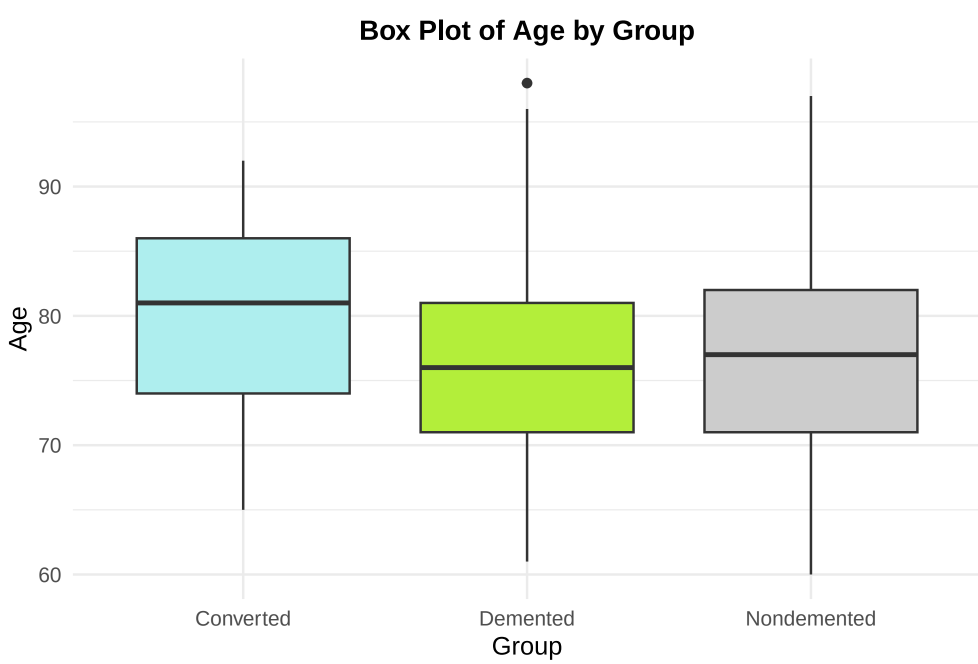
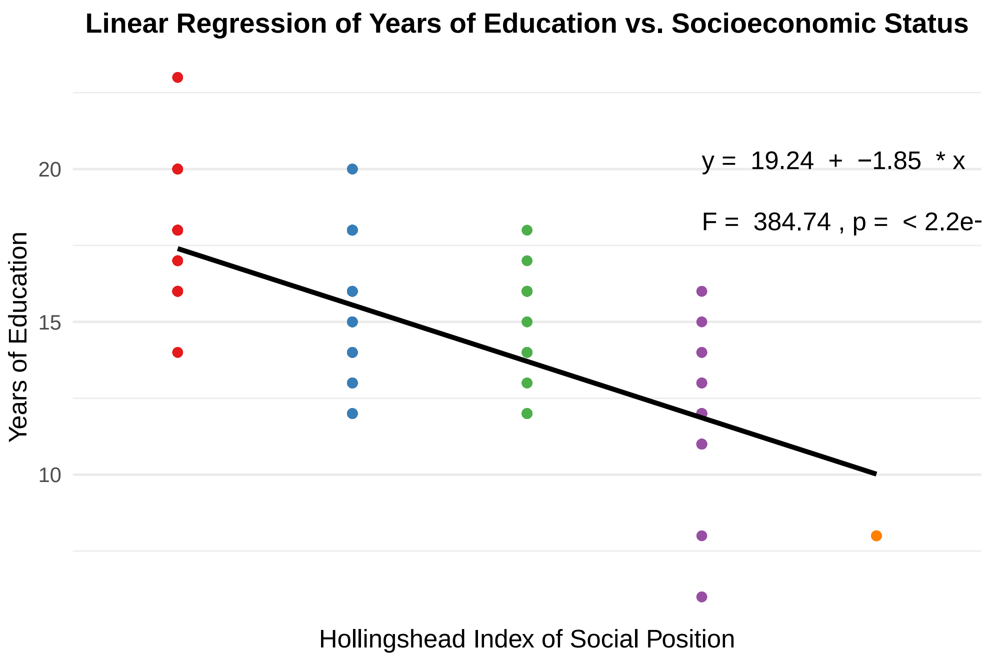


Table 1: Table 1: Tukey's HSD Results for Age vs. Group

	diff	lwr	upr	p adj
Demented-Converted	-3.4964828	-6.787092	-0.2058737	0.0342323
Nondemented-Converted	-2.6988620	-5.911511	0.5137872	0.1193746
Nondemented-Demented	0.7976208	-1.170014	2.7652553	0.6065021

Socioeconomic Status and Education

It was expected for years of education to decrease as Hollingshead Index of Social Position increases, because the status decreases with increasing Hollingshead Index of Social Position, where 1 is the highest status and 5 is the lowest. A linear regression was performed and as expected, the model was statistically significant with a p-value of 2.2e-16, showing that the predicted years of education decrease with decreasing socioeconomic status.



Age and CDR

It was expected for CDR to increase with age. The CDR is a 0-3 scale, with severe cognitive impairment being a 3. This dataset had CDR values ranging 0-2, with only 3 participants at a score of a 2. From the ANOVA with the Tukey's HSD test, there were no significant differences found in the age of each CDR rating. However, it would be interesting to redo this study and see if any differences arise when including more participants with a CDR of 2-3.

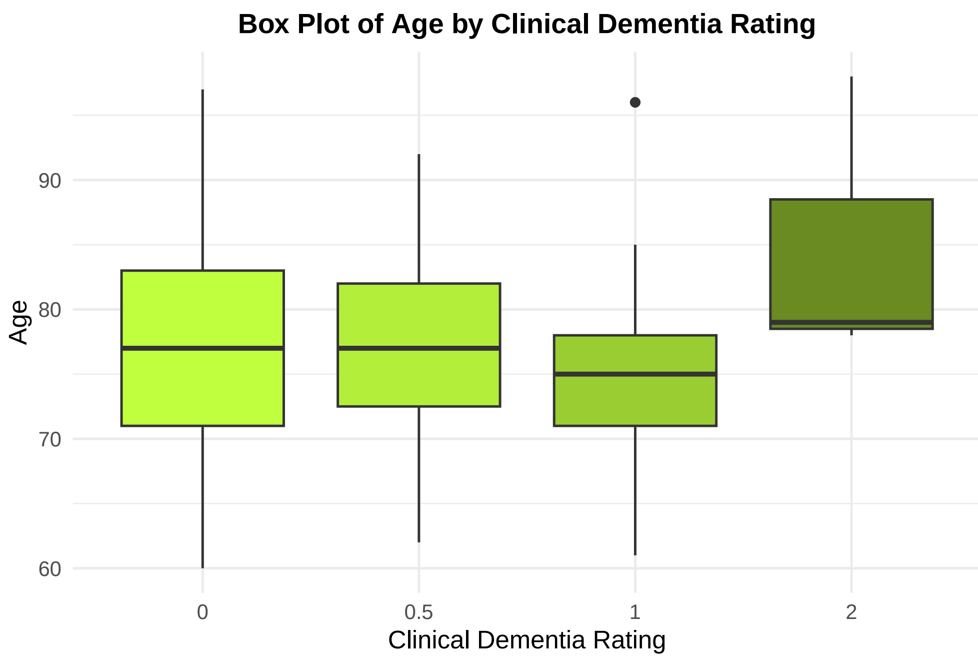


Table 2: Table 2: Tukey's HSD Results for Age vs. CDR

	diff	lwr	upr	p adj
0.5-0	0.226774	-2.005929	2.4594765	0.9936835
1-0	-2.545584	-5.896334	0.8051662	0.2049446
2-0	7.844660	-3.549911	19.2392313	0.2861884
1-0.5	-2.772358	-6.305788	0.7610725	0.1806789
2-0.5	7.617886	-3.831736	19.0675087	0.3162092
2-1	10.390244	-1.328818	22.1093062	0.1026389

Group and MMSE

It was expected that MMSE would be lower in the demented group because the MMSE assesses cognitive impairment, where lower scores indicate more significant cognitive impairment. Through an ANOVA with Tukey's HSD test, it was found that the demented group was significantly different than both the converted and non-demented group with a p-value of 0. An interesting finding from this analysis is that the converted and non-demented group are not significantly different. This may be explained by higher MMSE scores due to patients being in the early stages of dementia and cognitive impairment, or that the MMSE was performed while these patients were still considered non-demented.

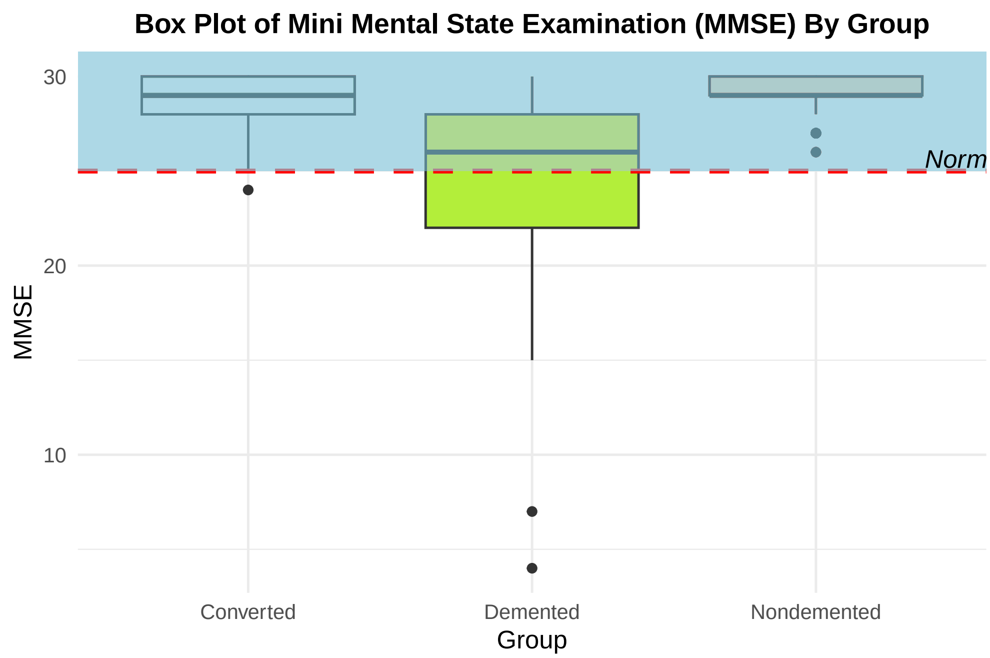


Table 3: Table 3: Tukey's HSD Results for MMSE vs. Group

	diff	lwr	upr	p adj
Demented-Converted	-4.1617868	-5.4261937	-2.897380	0.0000000
Nondemented-Converted	0.5506401	-0.6820813	1.783362	0.5451059
Nondemented-Demented	4.7124269	3.9544694	5.470384	0.0000000

nWBV And Age By Group

nWBV was shown to decline significantly with increasing age in previous studies, so this was analyzed separately by dementia group. All three groups had a linear correlation between nWBV and age, which confirmed that nWBV decreased with increasing age. The rates of decline were slightly different. The nondemented group showed a rate of decline at 0.0028 per year, whereas the converted group showed a 35% higher rate of decline than the nondemented group (0.0038 per year). This may be due to the converted group being in a more active cognitive decline. Interestingly, the demented group showed a lower rate of decline than both groups at 0.0019 per year, which may be due to that group having a lower nWBV to start. An interesting future analysis would be to reanalyze this data by percentage instead of raw value, to see if a different result shows for the demented group.

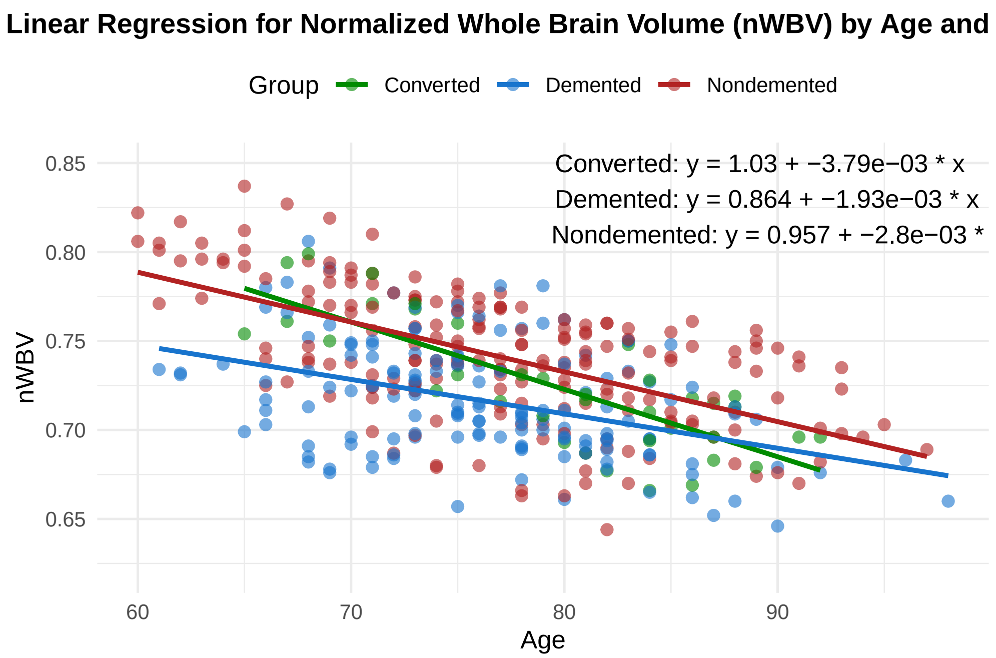


Table 4: Table 4: Linear Regression Results By Group

Group	Intercept	Slope	F_Statistic	P_Value
Non-demented	0.9565345	-0.0027983	106.22851	0e+00
Demented	0.8636706	-0.0019325	30.86854	1e-07
Converted	1.0256462	-0.0037855	59.43262	0e+00

Conclusion

Significant Results

The age of the converted group was significantly different than the age of the demented group with a p-value of 0.034, but no other comparisons from that analysis were significant. Years of education do in fact decrease as Hollingshead Index of Social Position increases, as the model obtained through linear regression was statistically significant with a p-value of 2.2e-16. There were no significant differences in age by clinical dementia rating, but it should be noted that there was an underrepresentation of patients with a CDR of 2-3. It was found that the MMSE score of the demented group was significantly different than both the converted and non-demented group with a p-value of 0. The converted and non-demented group are not significantly different in terms of MMSE score. All three groups (nondemented, demented, converted) had a negative linear correlation between nWBV and age, which confirmed that nWBV decreased with increasing age. Of the three groups, the converted group showed a the largest rate of decline, and the demented group showed the smallest rate of decline.

Future Directions

It would be interesting to redo this study and see if any differences in age arise when including more participants with a CDR of 2-3. The normalized whole brain volume data vs. age analysis by group should be redone by using a percentage or normalizing the y-axis due to the demented group's lower starting nWBV to see if there is a larger rate of decline.

Sources

Note: This data was obtained from the Open Access Series of Imaging Studies (OASIS), which is comprised by data made available from the Washington University Alzheimer's Disease Research Center, Dr. Randy Buckner at the Howard Hughes Medical Institute (HHMI) at Harvard University, the Neuroinformatics Research Group (NRG) at Washington University School of Medicine, and the Biomedical Informatics Research Network (BIRN).

Link to data: https://www.kaggle.com/datasets/ninadaithal/imagesoasis